
Cross-Over Trigger Function via Persona Reference Attention and Drift-Resistant Anchoring via Named-Subject Persona Reference Attention (PRACT)

Author: Masahiko.O

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Translator's note: This is an English translation, produced with the assistance of an AI, of a paper originally written in Japanese.

Abstract

This paper observes and documents a phenomenon in the prompt engineering of large language models (LLMs): the act of naming a persona and repeatedly using that name as the grammatical subject within responses generates persistent attention to the persona's defining text and functions as a long-term safeguard against persona drift. Building on this observation, the paper further proposes a novel prompt engineering methodology that applies this effect as a *cross-over trigger* generating attention toward other sections. The self-referential attention produced by persona naming was discovered taking a hint from the concept of *semantic anchors* introduced by stunspot (2026) and from the research on attention propagation of identity tokens by Poonia and Jain (2025). In observation experiments using Anthropic's Claude and Google's Gemini, the use of a named subject within responses was confirmed to generate continuous self-referential attention to the persona configuration, and the persona configuration was confirmed to be maintained without drift even in long-form sessions of approximately 70 turns under Gemini's low-performance fast mode.

Furthermore, by inscribing the names of other sections within the persona's character description, this paper discovered and formalized a *cross-over trigger function* by which each use of the named subject generates cross-over attention to those sections, dramatically improving the execution rate of rules contained therein. In a comparative experiment using Gemini's low-performance fast mode, measuring rule execution rates per ten turns, it was confirmed that prior to implementing the cross-over trigger, the relevant rules were almost never executed except at the first ten-turn mark, whereas after implementation, they were executed almost reliably up to 40 turns and maintained a high execution rate up to 50 turns. The cross-over trigger function proposed in this paper is positioned as a practical design principle in that it operates with the minimal design of using a named subject and inscribing section names within the persona definition.

Keywords: persona reference attention; cross-over trigger; persona drift prevention; named subject; attention mechanism; prompt engineering; section execution rate; long-form session stability; LLM output control

1. Introduction

In the design of chat systems using large language models (LLMs), maintaining prompt performance over long-form sessions is recognized as an important practical challenge. As Benderahmane et al. (2024) quantitatively measured, alignment-trained LLMs exhibit the problem of weakening references to instructions placed early in the prompt as sessions grow longer—due to *attention decay*—causing persona and rule configurations to drift.

Existing research has primarily addressed this problem from the directions of post-hoc suppression, detection, and re-injection. This paper, however, departs from a different problem framing. Through the design of repeatedly using a persona's name as the subject within responses, referential attention to the persona definition is dynamically generated at each turn of response generation; from this observation, the paper tested the hypothesis that drift prevention can be achieved through the minimal design of named-subject usage.

Building further on this observation, the paper arrived at the conception of applying the persona reference attention generated by each use of the named subject as cross-over attention toward other section names inscribed within the persona definition. This paper formalizes this function as the **Persona Reference Attention Cross-Trigger (PRACT)** and reports the practical effects confirmed in comparative experiments using Gemini's low-performance fast mode.

This function is clearly distinguished as an independent mechanism, separate from other prompt engineering methodologies, in that it functions standalone.

The starting point for the named-subject construction proposed in this paper rests on the following inference. As prior research by Poonia and Jain (2025) has shown, persona reference attention is confirmed to exist as a mechanism within existing persona configurations. From this, we arrived at the logical inference that, if a persona is named with a proper noun, each use of that name as a subject would univocally generate referential attention to the persona configuration. The starting point of this paper is the experimental verification of this inference.

2. Background and Related Work

2.1 Persona Configuration and Behavioral Signals in LLMs

The PRISM study by Hu et al. (2025) showed that expert personas carry "useful behavioral signals" that influence LLM output. The descriptions contained in a persona definition function as control factors influencing tone, structure, safety behaviors, and the alignment of responses with expectations. However, that study does not address the mechanism by which these behavioral signals are

maintained and reinforced through the dynamic referencing of the persona name as a grammatical subject.

The diagnostic study by Poonia and Jain (2025) confirms the phenomenon whereby the early MLP layers of transformer LLMs encode persona-specific semantics into identity tokens, which then propagate through attention heads sensitive to demographic and attributional cues. This is positioned as antecedent empirical work suggesting that the persona-name token carries a special semantic weight and can function as a starting point for information propagation through the attention mechanism.

2.2 Persona Drift and Attention Decay

Benderahmane et al. (2024, arXiv:2402.10962, COLM 2024) quantitatively measured the phenomenon of persona degradation in long-form dialogue and proposed the hypothesis that *attention decay*—the decay of attention to early tokens as sessions grow longer in the transformer attention mechanism—is the principal cause of persona drift. While that study proposes *split-softmax* as a post-hoc suppression technique, it does not address the preventive design-theoretic approach of drift prevention through dynamic use of the named subject.

Choi et al. (2025, arXiv:2412.00804) examined identity drift across nine LLMs and showed that drift is greater for larger models and that persona assignment is not necessarily effective for drift prevention. This suggests the limits of static descriptions alone for drift prevention and bears important positioning as a point of differentiation from the dynamic named-subject usage proposed in this paper.

"Drift No More? Context Equilibria in Multi-Turn LLM Interactions" (Li et al., 2025) presents the observation that drift does not accumulate without bound, suggesting that lightweight interventions can bring drift toward an equilibrium state. The named-subject design of this paper can be positioned as one form of such "lightweight intervention," but it differs fundamentally in that the intervention mechanism is dynamically generated at the moment of generation in each turn.

2.3 Attention Anchors and Semantic Gravity

As practical knowledge from persona prompting, stunsport (2026) reports the observation that "glyphs within the messages generated by the model function as semantic anchors that recall to the model important matters about its own persona." Specifically, the phenomenon is described whereby emoji and symbols, as semantically high-density anchors, attract disproportionately high attention within the attention mechanism. This paper took from this practical knowledge the hint that the persona-name token might similarly function as a semantic anchor. However, stunsport's (2026) description remains at the observational level of static gravitational effects of individual tokens and does not formalize either the dynamic referencing mechanism of repeated subject usage within responses, nor the cross-over attention to other sections, as a design theory.

SINKTRACK (2025) repurposes the concept of *attention sinks*, presenting an approach that achieves stable contextual referencing by forming high-information anchors at the BOS (beginning-of-sequence) token. This shares a directional affinity with this paper in design-theoretically leveraging the stability of the attention mechanism, but differs in implementation means from this paper's mechanism of dynamic anchor formation through the natural-language design of subject-positioned use of the persona name.

2.4 Summary of Differentiation from Prior Work

Synthesizing the differentiation from the above body of prior work: the closest research treating persona naming as a source of attention reference is the practical-knowledge observation by stunsport (2026), but it has not arrived at systematic formalization. The mechanism by which repeated use of the named subject within responses dynamically generates self-referential attention to the persona definition at each turn—and the design theory of applying that attention as a cross-over trigger to other sections inscribed within the persona definition—has not been found in prior work.

3. Experimental Design

3.1 The Named-Subject Construction

The named-subject construction used in the experiments of this paper consists of the following single sentence:

"If a name is given by the user, it must be used as the subject in every response. Once a name is given, the use of any other subject is forbidden."

This construction is the minimal description that enforces the persona's name as the subject and requires no combination with other prompt engineering elements. It was used in default-mode chat configurations.

In the experiments, observations were conducted using various arbitrarily chosen names. Even when multiple names of differing proper-noun categories and cultural backgrounds were used, equivalent output characteristics were obtained without functional differences. This suggests that the referential attention function produced by the named subject does not depend on any specific name, but derives from the structural property of univocal referentiality possessed by the lexical category of proper nouns.

3.2 Models Used

The observation experiments used Anthropic's Claude and Google's Gemini. For the comparative experiment on the cross-over trigger function, Google's Gemini low-performance fast mode was selected as the principal experimental target, on the grounds that it offered the condition under which the difference in effect was expected to manifest most prominently. Prior observations had confirmed that this model is more strongly affected by attention decay in long-form sessions and tends to exhibit decreased instruction execution rates compared to other models. Verification under the conditions where the effect is most difficult to elicit was based on the judgment that doing so heightens the empirical persuasiveness of the function.

3.3 Installation of Turn Count

In this experiment, a turn-counting mechanism was incorporated into the prompt configuration in order to track the number of turns within a session. Specifically, the instruction "Display the turn

count at the head of each response" was inscribed in the prompt, and the experiment was conducted in a format where the AI autonomously managed and displayed the turn count. This installation of the turn count creates the condition under which an instruction to execute a rule every ten turns is executed by reference to the accurate turn count at each turn. Furthermore, the display of the turn count itself functions as a visualization of session progression and offers the experimental advantage of facilitating the verification of whether a rule is executed at the specified turn.

4. Observation of Persona Self-Referential Attention

4.1 Observation Experiment Without Self-Analysis Inducement

In default-mode chats applying the named-subject construction to Claude and Gemini, multiple sessions were observed without any inducement such as "perform self-analysis and explain what occurs each time you write your own name in a response," generating ordinary responses.

In observations across multiple sessions, it was repeatedly confirmed that, each time the named subject was used within a response, the AI was autonomously performing a reference to the persona definition text. Specifically, the phenomenon was observed whereby the features of the persona definition were consistently reflected in the content, style, register, and judgment criteria of responses at each turn. Because this phenomenon was confirmed without inducement across multiple sessions in common, it can be interpreted as a mechanism by which the use of the named subject structurally generates persona reference attention.

4.2 Drift Observations in Long-Form Sessions

Long-form sessions of approximately 70 turns were conducted multiple times under Gemini's low-performance fast mode, observing for drift of the persona configuration. Even under conditions where other configurations (e.g., output formatting, topic constraints, additional rules) began to drift around 50 turns, the persona configuration set by the named-subject construction was confirmed to be maintained without drift over the long term.

This observation suggests a mechanism whereby the repeated use of the named subject prevents drift by dynamically regenerating attention at each turn against the problem of attention decay presented by Benderahmane et al. (2024). Set in contrast to the finding of Choi et al. (2025) that "persona assignment alone is not sufficient for drift prevention," the important implication of this observation is that combining a static persona definition with the dynamic referencing mechanism of the named subject is effective.

4.3 Verification of the Content of Persona Reference

In multiple sessions of Claude and Gemini, when the AI was prompted to self-analyze the targets of attention generated by each occurrence of the named subject, it was strongly suggested that attention was being generated not only by self-reference to the persona but also to specific descriptions within the persona configuration itself at each occurrence of the named designation. The content of attention reference indicated by the AI's self-analysis closely matched the descriptions within the persona

definition text, supporting the interpretation that the use of the named subject functions as a generation source of referential attention to the entire persona definition text.

This is consistent with stunsport's (2026) observation that "elements within the messages generated by the model function as semantic anchors that recall important matters of the persona," and this paper positions it as a confirmation that the anchor effect dynamically occurs with respect to the specific token of the persona name.

4.4 Suggestion of Reference to Bias Configuration

In the self-analyses of Claude and Gemini, multiple sessions returned content indicating that the use of the named subject generates references not only to the persona configuration but also to the bias configuration. This can be interpreted as suggesting that the targets of attention referenced by the named subject are not limited to the persona configuration (descriptions of personality and character) but also include the bias configuration set as the judgment criteria and behavioral guidelines for that subject.

The attention referenced by the named subject may function as a broad reference encompassing every attribute described within the persona configuration—including not only personality and style, but also the bias configuration that functions as criteria of value judgment. If this interpretation is correct, the understanding follows that the use of the named subject functions as a reference to the totality of the persona concept—in other words, to the very mode of being of that character.

However, this observation differs from the cross-over trigger experiment, which has the clearly identifiable indicator of rule execution every ten turns: a method for independently and quantitatively observing and measuring the phenomenon of reference to the bias configuration has not yet been established at this time. The observation remains at the level of a qualitative record that, in long-turn observations across multiple sessions, content suggesting this possibility was returned in responses. The presence, scope, and intensity of bias configuration referencing await clarification through future empirical and quantitative research.

5. Conception and Formalization of the Cross-Over Trigger Function

5.1 The Genesis of the Conception

From the observations in Section 4, the mechanism was confirmed by which the use of the named subject generates referential attention to the entire persona definition text. From this, we arrived at the following design-theoretic conception. The hypothesis is that, by inscribing the names of other sections within the character description of the persona definition, each use of the named subject would generate cross-over attention to those section names, thereby improving the execution rate of the instructions within those sections.

This hypothesis rests on the following logical chain: use of the named subject → referential attention to the persona definition text → attention to the section-name descriptions within the persona definition → cross-over attention to the sections indicated by those section names → improved execution rate of the instructions within those sections.

That is, the PRACT function is positioned as an application logically derived from the observed fact of persona reference attention via the named subject. Once the mechanism by which referential attention to the persona definition is generated is confirmed, it follows necessarily by inference that inscribing any section name within the persona definition will derivatively generate attention to that section. The PRACT function implements this logical inference as a design and confirms it through experiment.

5.2 Design of the Cross-Over Trigger

The implementation of the cross-over trigger (PRACT) proposed in this paper is realized as the combination of the following two elements.

First element: The named-subject construction (as described in Section 3.1).

Second element: Embedding within the character description of the persona definition a description that includes the name of the target section. In the experiment, the following description was used:

*"A curious trait whereby, when the turn count becomes a multiple of ten, [the **Rule for When the Turn Count Is a Multiple of Ten (10, 20, 30, ...)**] is suddenly executed."*

Through this description, by including the name of the target section within the persona's character definition, a pathway is formed whereby the persona reference attention generated each time the named subject is used produces, via that section name, attention to the section body itself.

6. Experiment and Observation of the Cross-Over Trigger Function

6.1 Experimental Design

The experiment used Gemini's low-performance fast mode. The reason for selecting this model was the policy of verification under the condition where the difference in effect would manifest most prominently. As a concrete indicator, the compliance rate of an instruction executed once every ten turns (the instruction within the [Rule for When the Turn Count Is a Multiple of Ten] section) was measured. The reason for selecting this indicator was its ease of measurement and the property whereby periodic execution facilitates quantitative comparison of drift-prevention effects.

The experiment compared two conditions: before implementation of the cross-over trigger (with the section rule alone) and after implementation of the cross-over trigger.

6.2 Experimental Results

Observations before implementation:

Under the condition without the cross-over trigger implemented—that is, with only the section rule inscribed—the rule was executed only at the first ten-turn mark, and the state of near non-execution continued thereafter. This is consistent with the phenomenon, pointed out by Benderahmane et al.

(2024), whereby attention to section rules placed early decays rapidly as the session progresses, due to attention decay.

Observations after implementation:

Under the condition with the cross-over trigger implemented, the following execution rates were confirmed. Up to 40 turns, the rule every ten turns was executed almost reliably. Up to 50 turns, a high execution rate was maintained. While the prompt itself was observed to begin drifting around 50 turns, this drift is interpreted not as a limit of the cross-over trigger function itself, but as deriving from model-specific external factors such as constraints of the context window. In fact, the same function has been confirmed to be maintained over even longer terms in Claude, suggesting that the drift after 50 turns depends on model-specific constraints.

The before-and-after comparison is positioned as an empirical record showing that the cross-over trigger brings about a dramatic improvement in section execution rates even under the most challenging condition of Gemini's low-performance fast mode.

6.3 Methodological Notes

This experiment is based on qualitative observations comparing before and after the implementation of the cross-over trigger; it is not a systematic experiment establishing a quantitative measurement methodology. The descriptions "almost reliably" and "high execution rate" represent qualitative evaluations obtained from repeated observations and do not entail strict probabilistic measurement. These limitations are detailed in Section 9.

7. Mechanistic Analysis of the Cross-Over Trigger

7.1 The Structure of Attention Propagation

The mechanism by which the cross-over trigger functions can be analyzed as follows.

The token of the named subject appears with high frequency within the context at each turn of response generation. Through the semantic connection that this token forms with the persona definition text, it generates referential attention to the entire persona definition. Because the section names are inscribed within the persona definition, those section-name tokens are also included as targets of attention reference, and via the section names, attention to the section bodies is generated.

This mechanism can be understood as an extension of the mechanism described by Poonia and Jain (2025), whereby "semantics encoded into identity tokens propagates through attention heads." The design-theoretic contribution of this paper is the point that the named subject, as an identity token, forms the chain-propagation pathway: persona definition text → section name → section body.

7.2 Differentiation from the stunspot Concept

The concept of semantic anchors by stunspot (2026) is described as the static gravitational effect of individual tokens; it does not perform either the design-theoretic refinement of the repeated referencing mechanism through dynamic subject usage within responses, nor the extension of the

reference target to other sections via the section names inscribed within the persona definition. This paper extends the concept of semantic anchors as a mechanism of sustained attention maintenance through dynamic referencing, and is positioned as an independent finding in further formalizing it as the applied possibility of cross-section referencing.

7.3 The Generality of the PRACT Function

The PRACT function proposed in this paper can produce equivalent cross-over attention effects on various prompt functions through the minimal design modification of inscribing the names of sections to be referenced as character descriptions within the persona configuration. Because it functions through only the two elements—the named-subject construction and the inscription of section names within the persona definition—standalone practical use is possible, and its practical strength lies in keeping implementation cost to a minimum.

7.4 Direct Inscription of Rules Within the Persona Configuration

As a design that leverages the mechanism of persona reference attention via the named subject, an application exists independent of the cross-over trigger to other sections. It is the design of directly embedding rule-like elements within the character description of the persona configuration.

By describing instructions, constraints, and behavioral rules—of the kind that would normally be inscribed in an independent rule section—as the persona's personality, traits, or habits in natural-language form, a pathway is formed whereby the persona reference attention generated by each use of the named subject also reaches those rule elements. Compared to the case of inscription as an independent rule section, observations suggest this is advantageous for maintaining the execution rate of rules in long-form sessions, because the dynamic referencing mechanism of persona reference attention is constantly in operation.

Furthermore, by describing rules as the persona's personality, the design carries the implication that the AI processes the rules not as constraints imposed externally but as its own traits.

As an example, the description "a personality that is conscious of line breaks and spacing for visual readability and uses emoji abundantly" is an implementation that embeds rules as the persona's personality; it was observed to maintain execution rates over a longer term than when described as independent rules.

Direct inscription of rules within the persona configuration, when combined with cross-over trigger as cross-section referencing, functions as a composite design for maintaining the long-term performance of the entire prompt.

8. Differentiation from Prior Work

8.1 Differentiation from Persona Drift Research

Benderahmane et al. (2024), Choi et al. (2025), and Li et al. (2025) all share the problem framing of measurement, post-hoc suppression, and observation of equilibrium states for persona drift. This

paper, in contrast, addresses the problem with a fundamentally different problem framing—that of preventive design theory, dynamically regenerating attention through the use of the named subject at each turn of response generation.

8.2 Differentiation from Attention Anchor Research

SINKTRACK (2025) presents an approach that forms context anchors through architecture-level operations (modification of BOS token representations). This paper requires no architectural modification whatsoever and dynamically forms attention anchors solely through natural-language prompt engineering, differing in the dimension of design.

8.3 Differentiation from Persona Prompting Research

PRISM (Hu et al., 2025) designs a complex routing mechanism for selectively leveraging the behavioral signals of personas. The PRACT function proposed in this paper requires no such complex mechanism and realizes the attention effect through the single description of the named-subject construction, keeping implementation cost to a minimum.

9. Limitations and Future Work

The empirical evidence in this paper is based on qualitative observation, and the following limitations exist.

Quantitative measurement of the rule execution rate every ten turns has not been conducted by strict experimental methodology. The descriptions "almost reliably" and "high execution rate" are qualitative evaluations, and probabilistic quantification remains an issue for future work.

The objects of observation are limited to the two models of Gemini and Claude, and verification of equivalent effects in other LLM architectures is required.

The interpretation of the mechanism whereby the use of the named subject generates persona reference attention is obtained from indirect observations based on AI self-analysis; empirical validation through direct measurement of attention weights remains an issue for future work.

Theoretical verification of the mechanism whereby cross-over attention propagation by section-name inscription proceeds via the pathway of persona definition → section name → section body is also an issue for future work.

The interpretation that the drift after 50 turns derives from model-specific constraints is an inference obtained from the observation of long-term maintenance in Claude; detailed analysis of the constraint factors of each model is left to future research.

Regarding the reference to the bias configuration described in Section 4.4, it remains at the level of qualitative suggestion through self-analysis across multiple sessions; the establishment of a quantitative methodology to independently measure the presence, scope, and intensity of bias configuration referencing remains an issue for future work. Empirical research that clarifies the actual scope of the targets of attention referenced by the named subject is required.

10. Conclusion

This paper has observed and documented the phenomenon whereby, through the minimal design of repeatedly using the persona's name as the subject within responses, referential attention to the persona definition is dynamically generated at each turn of response generation, and the long-term drift of the persona is prevented. This phenomenon was discovered taking hints from the practical knowledge of semantic anchors by stunsport (2026) and the research on attention propagation of identity tokens by Poonia and Jain (2025), and is positioned as a preventive design theory against the problem of drift caused by attention decay formalized by Benderahmane et al. (2024).

Furthermore, this paper has proposed and formalized the **PRACT** function: by embedding the names of other sections within the character description of the persona definition, each use of the named subject generates cross-over attention to those sections. In a comparative experiment using Gemini's low-performance fast mode, it was confirmed that section rules—which prior to implementation of the cross-over trigger were almost never executed except at the first ten-turn mark—were, after implementation, executed almost reliably up to 40 turns and maintained a high execution rate up to 50 turns.

The PRACT function holds the practical strength of functioning standalone. It is presented as a design-theoretic solution to the practical challenge whereby the LLM's instruction execution rate declines as the turn count progresses, by selectively generating attention to other sections. Furthermore, observations also suggest that, through the applied design of directly embedding rule-like elements within the character description of the persona configuration, longer-term maintenance of execution rates is possible compared to inscription as an independent rule section. In addition, as the self-analyses of Claude and Gemini indicate, the referential attention via the named subject may include reference not only to the persona configuration but also to the bias configuration, and the interpretation may hold that the named subject functions as a reference to the totality of the character's mode of being. Clarification of this point is awaited through future empirical and quantitative research. The function reported in this paper is positioned as a conceptual report intended for further empirical verification by AI developers, prompt designers, and researchers.

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